

Air Quality in Taiwan

...

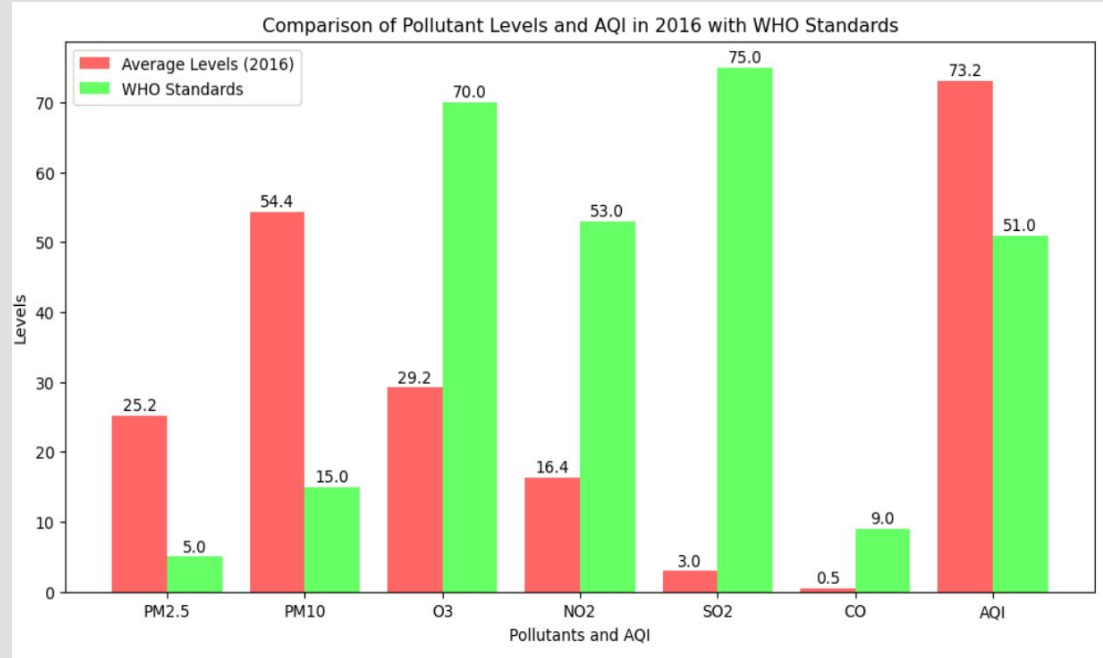
Griffin Munhall, Sarah Oakes, Daniel Mintz

Part 1: Background

...

Problem Statement:

Air pollution is a global problem that create severe risk to public health, various ecosystems, and the climate with almost 99% of the world's population is breathing air that exceeds WHO guidelines. This widespread issue demands mitigation of emissions from industrial, vehicular, and natural sources.



Dataset: Overview

- Dataset sourced from Kaggle.com, a popular machine learning website.
- File has 5.8 million rows and is 800 MB, making processing slower.
- It spans from 2016-2024 with longitude, latitude, and location data for regional and spatial analysis.
- Large size and time span also allow for exploration of long-term air quality trends across locations.

Dataset: Key Numerical Variables

PM2.5 & PM10

PM2.5: Fine particulate matter (2.5 micrometers), harmful to lungs and bloodstream.

PM10: Particulate matter (10 micrometers), affects respiratory health.

AQI

Air Quality Index (AQI) is a measure which shows the level of air pollution and its potential for adverse health effects. An increase in the air quality index signifies increasing health risk.

O3, NO2, SO2 & CO

O3: Harmful pollutant

NO2: Respiratory irritant.

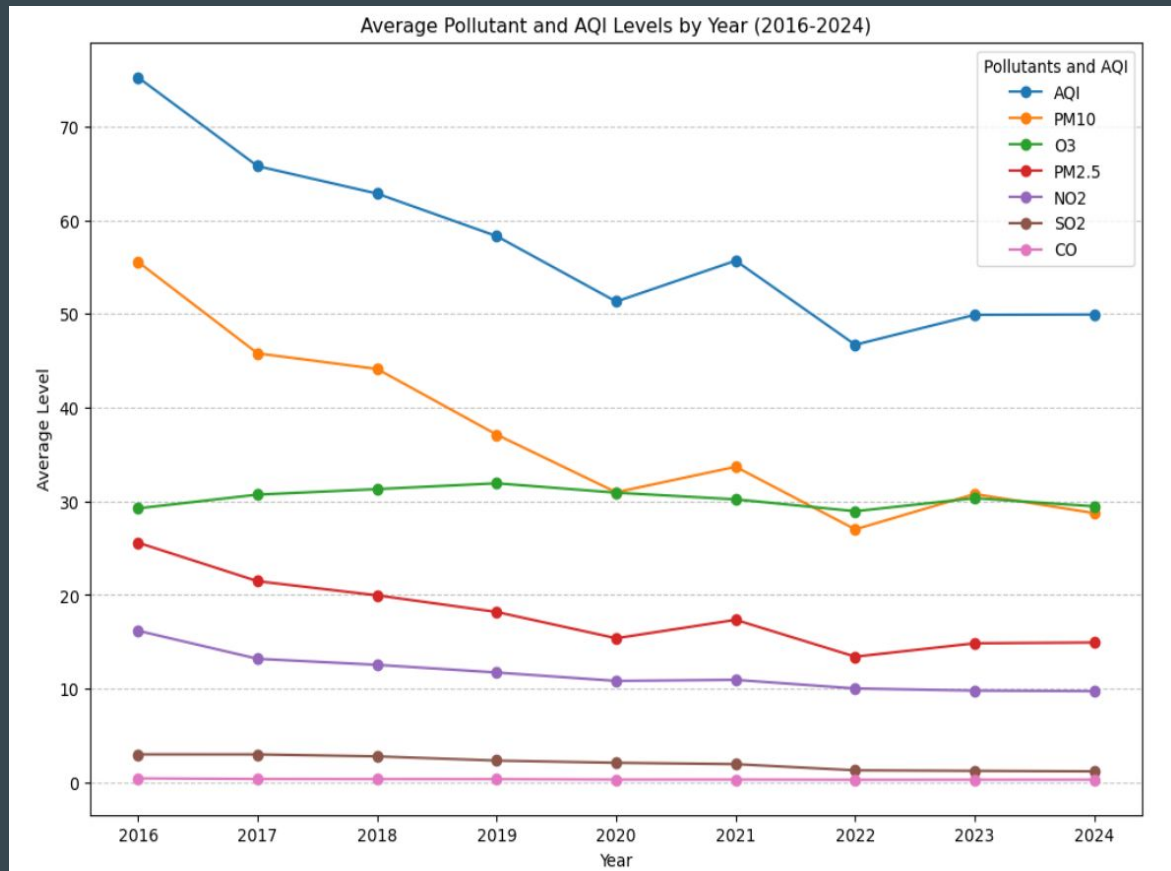
SO2: Respiratory and environmental harm.

CO: Reduces oxygen delivery.

Part 2: Analysis

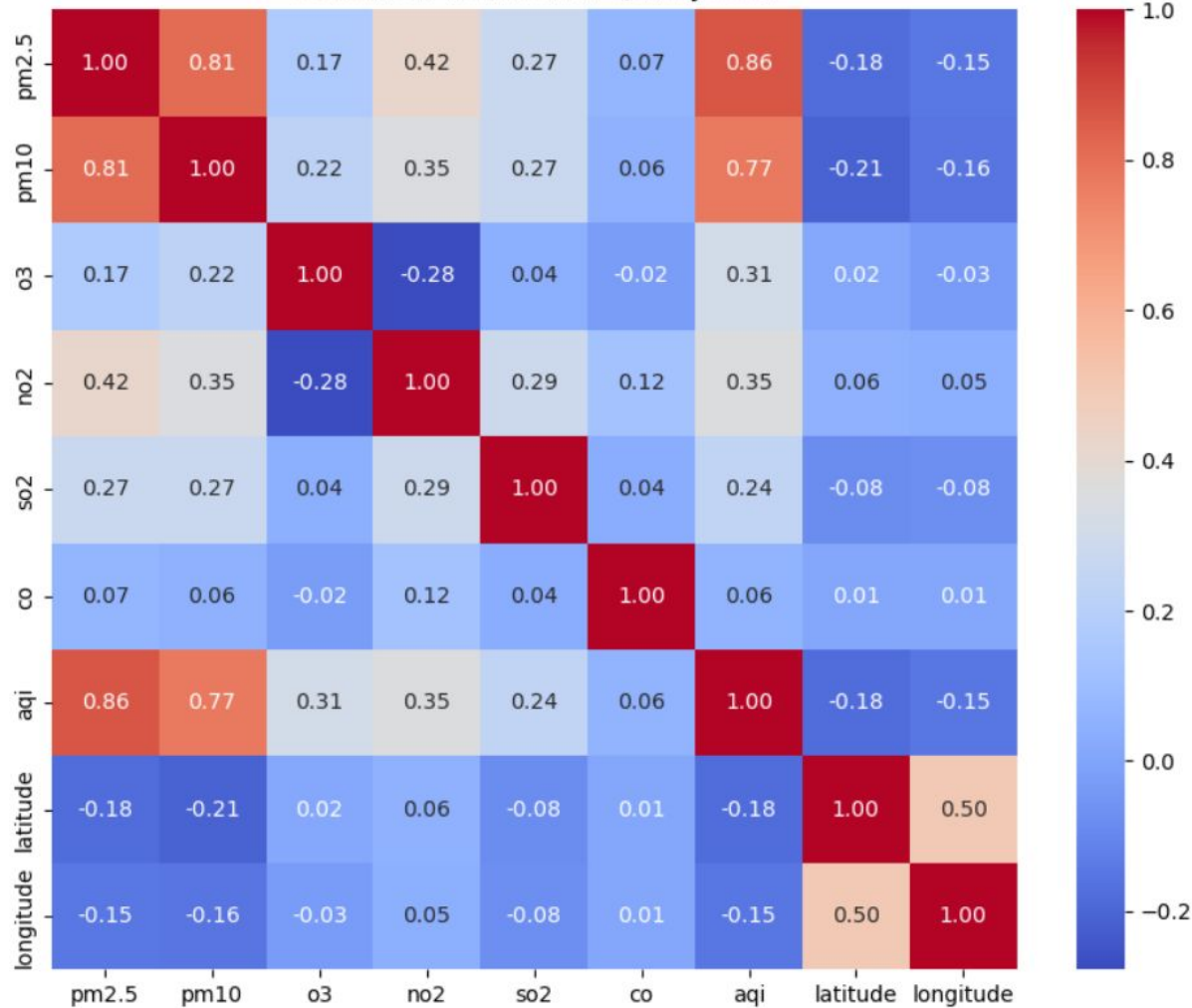
...

From 2016-2024: Temporal Analysis



- Distinct decrease in PM2.5, PM10, AQI
- Extremely similar lines between PM10 and AQI: Is there a correlation?

Correlation Matrix of Air Quality Data



PM2.5 and PM10 are strongly correlated with AQI in this dataset, at 0.86 and 0.77 respectively.

PM2.5 particles can penetrate deep into the lungs and enter the bloodstream, posing significant health risks, which is why PM2.5 is weighted heavier when calculating AQI.

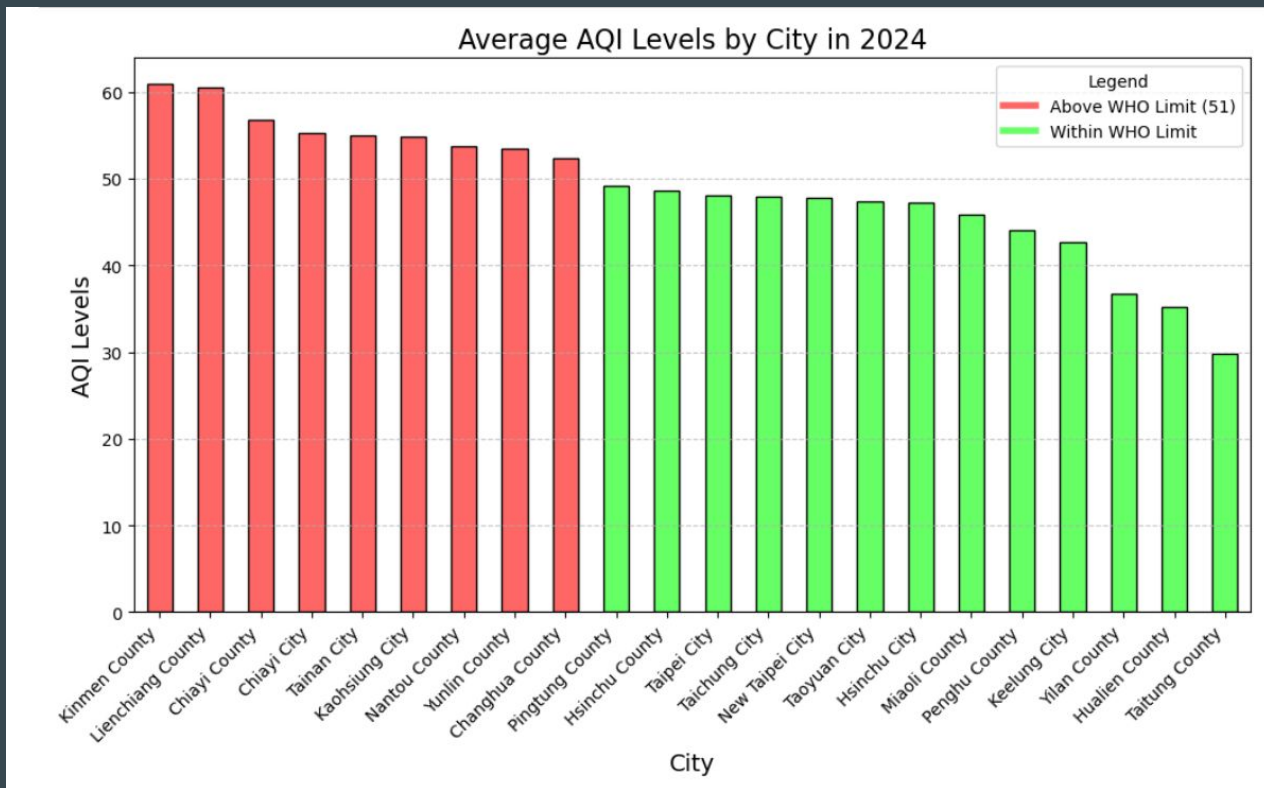
Why do PM2.5 and PM10 have high correlation to AQI?

- PM2.5 and PM10 are critical to the calculation of AQI, which assesses air pollution levels and their potential health impacts.
 - Areas with elevated PM2.5 levels often show higher AQI readings to reflect their increased danger.
-

What Factors Contribute to High Levels of PM?

- **Industrial Activity:** Cities like Kaohsiung and Tainan, represented by red bars, are major industrial hubs
 - **Geographical and Meteorological Conditions:** Southern regions are more affected by transboundary pollution from neighboring countries, especially during the northeast monsoon season.
-

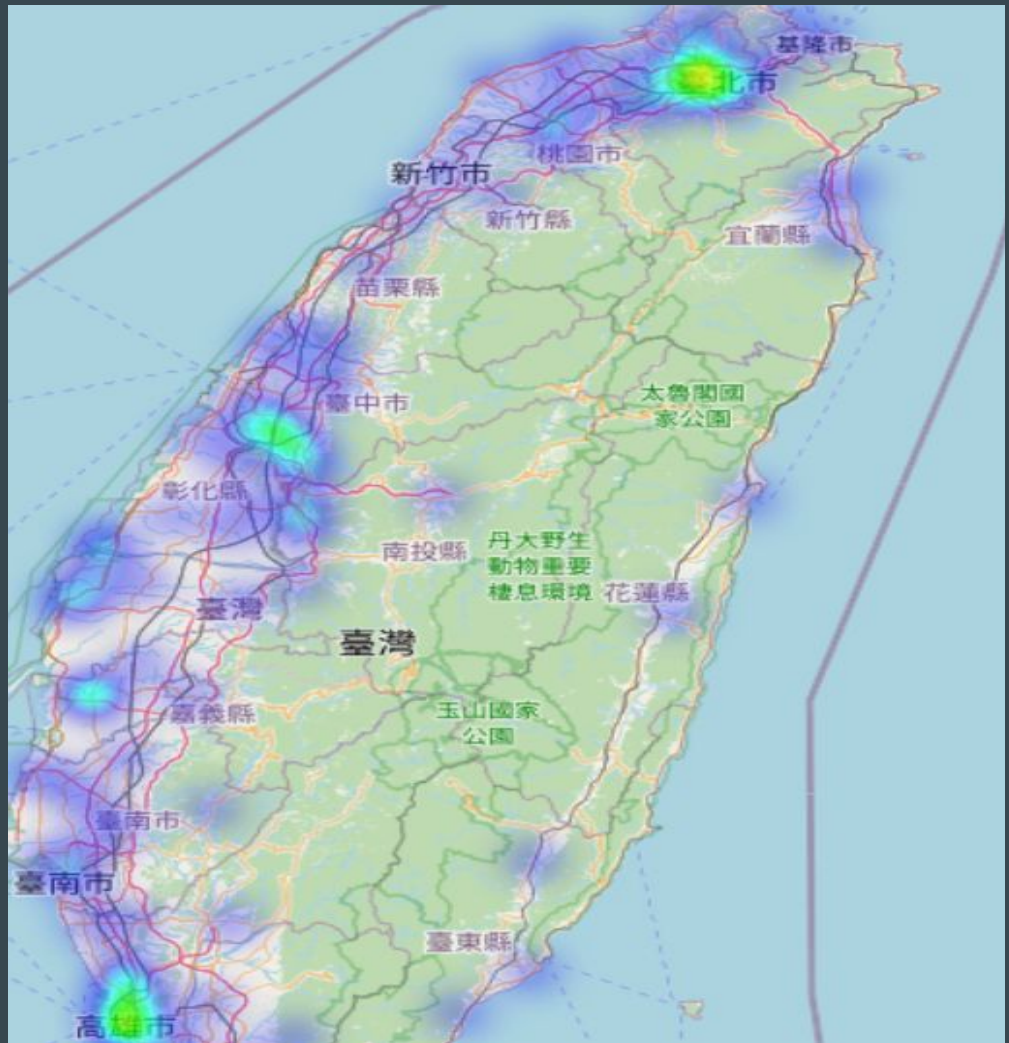
Spatial Analysis: Overview



Eastern Taiwan

Good Air Quality

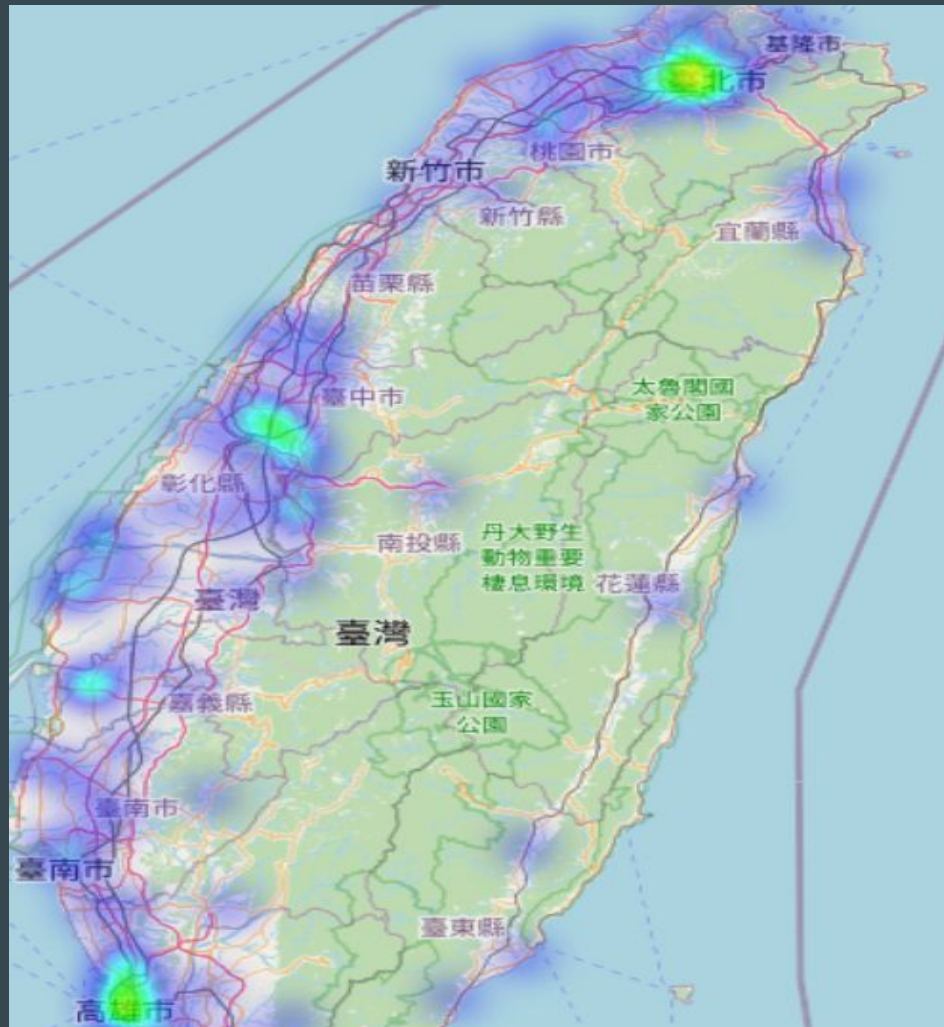
The eastern areas, having less industrialization and lower population density, have lower AQI levels, showing their better air quality.



Western Taiwan

Poor Air Quality

Major cities like Taipei, Taichung, and Kaohsiung, have high AQI levels. The concentration of industrial activities and large urban populations contribute significantly to pollutant levels.



Spatial Analysis: Influencing Factors

Industrial and Urban Emissions

The western areas increased industrial operations and vehicle emissions are the main contributors to higher pollution levels.

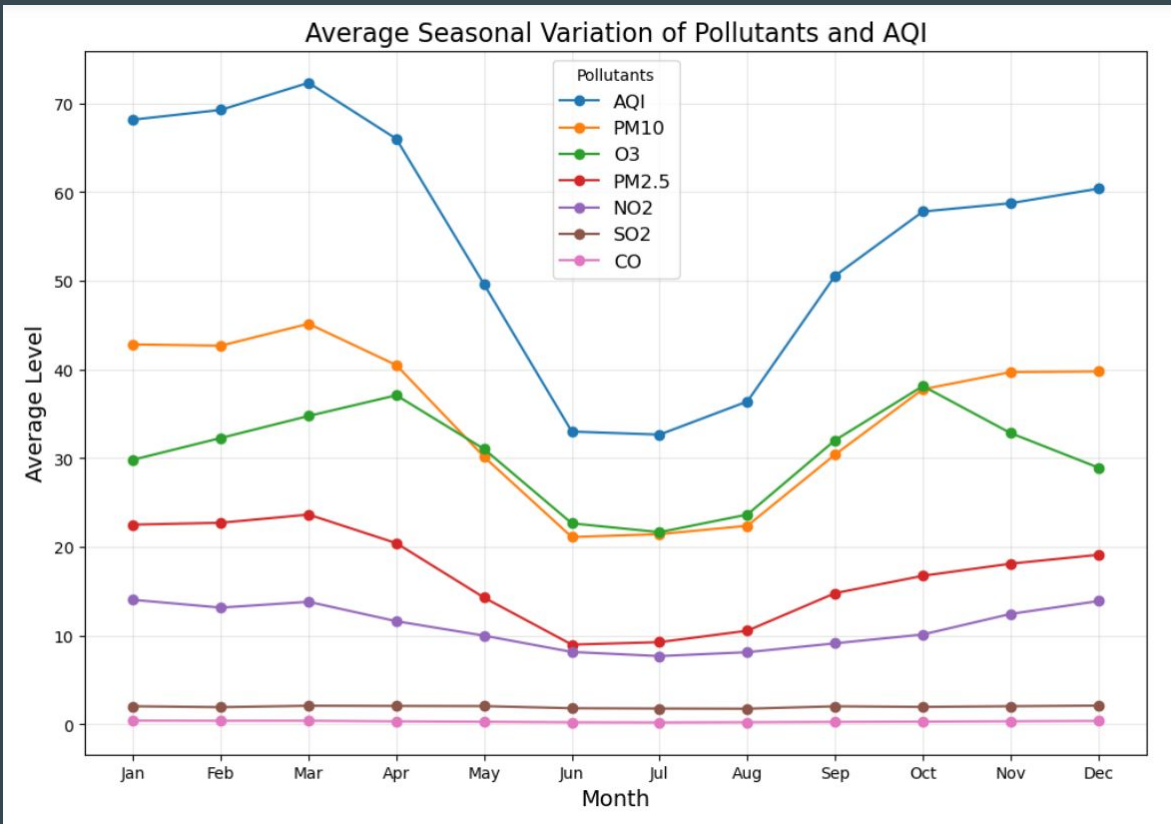
Topography and Meteorology

Taiwan's central mountain range acts as a barrier, limiting pollutants scattering from west to east. Wind patterns influence the collection and movement of pollutants, trapping them in western regions.

Transboundary Pollution

During certain seasons, pollutants from neighboring regions can move to Taiwan, making air quality issues worse, especially in the western region.

Seasonal Variation of Pollutants and AQI



- **Winter Months** : The temperature reduction traps pollutants near the surface, causing poor air quality.
- **Summer Months** : Improved air quality is due to mixing in the atmosphere and stronger winds that scatter pollutants more effectively.

Part 3: Conclusion

...

Current Taiwanese Initiatives

- The Air Pollution Control Action Plan (2020-2023) sets goals, such as achieving an average PM2.5 concentration of $15 \mu\text{g}/\text{m}^3$ by 2023. This goal was achieved ahead of schedule, with levels dropping from $16.2 \mu\text{g}/\text{m}^3$ in 2016 to $12.4 \mu\text{g}/\text{m}^3$ in 2022.
- The Air Pollution Control Program (2024-2027) is also moving Taiwan forward, aiming to reduce PM2.5 concentrations to $13.0 \mu\text{g}/\text{m}^3$ by 2027 and decrease the number of days with AQI over 150 by 80% compared to 2019.
 - By 2030: Electrify all government vehicles and public buses.
 - By 2040: 100% market share for electric cars and scooters.

Recommendations

- **Target Industrial Regions:** Prioritize funding and policies in counties like Kinmen and Chiayi, where AQI is over the WHO standards due.
- **Focus on PM Reduction:** Implement procedures and goals to reduce these pollutants, as they consistently breach WHO guidelines and correlate strongly with higher AQI.
- **Seasonal and Yearly Trends:** Address seasonal pollution spikes and maintain long-term improvements of pollutants like NO₂ and SO₂ over more recent years.

Questions?

...